

MTE 203 – Advanced Calculus

Week 3 Tutorial

Problem 1 [S. 11.10, P.5]

Find the parametric and vector representations for the curve given by the intersection of the surfaces

$$z + 2x^2 = 1$$

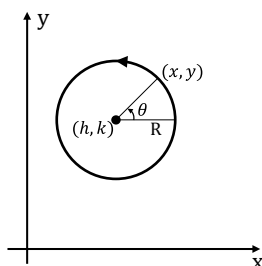
and

$$y = z$$

directed so that x decreases along the curve. Sketch the curve indicating its direction.

Problem 2 [S. 11.13, P.16]:

A particle travels counterclockwise around the circle $(x - h)^2 + (y - k)^2 = R^2$ drawn in the figure below:



- Show that the speed of the particle at any time is $|\vec{v}| = \omega R$ where $\omega = \frac{d\theta}{dt}$ is called the angular speed of the particle.
- Find the unit tangent vector to the curve at the point $P_0 = \left(\frac{R}{\sqrt{2}} + h, \frac{R}{\sqrt{2}} + k\right)$.
- Write a line equation for a tangent line passing through the point $P_0 = \left(\frac{R}{\sqrt{2}} + h, \frac{R}{\sqrt{2}} + k\right)$.